

Problem sheet 1

1. Let A, B and C be three arbitrary events. Using only the operations of union, intersection and complement, write down expressions for the events that, of A, B, C ;
 - a. Only A occurs.
 - b. Both A and B occur, but C does not.
 - c. All three events occur.
 - d. At least one event occurs.
 - e. At least two events occur.
 - f. One and only one event occurs.
 - g. Exactly two events occur.
 - h. No events occur.
 - i. Not more than two events occur.

$$a. A \cap \bar{B} \cap \bar{C} = A \cap \overline{B \cup C}$$

$$b. A \cap B \cap \bar{C}$$

$$c. A \cap B \cap C$$

$$d. S - \overline{A \cup B \cup C} = A \cup B \cup C$$

$$e. (A \cap B) \cup (A \cap C) \cup (B \cap C)$$

$$f. (A \cap \overline{B \cup C}) \cup (B \cap \overline{A \cup C}) \cup (C \cap \overline{A \cup B})$$

$$= A \cup B \cup C - (A \cap B) \cup (A \cap C) \cup (B \cap C)$$

$$g. \underline{(A \cap B) \cup (A \cap C) \cup (B \cap C) - A \cap B \cap C}$$

$$(\bar{A} \cap B \cap C) \cup (A \cap \bar{B} \cap C) \cup (A \cap B \cap \bar{C})$$

$$h. \overline{A \cup B \cup C}$$

$$i. S - A \cap B \cap C = \underline{\overline{A \cup B \cup C} + A \cup B \cup C} - A \cap B \cap C \quad \overline{A \cap B \cap C}$$

2. A fair-sided die is thrown twice.

- Write down the sample space for this experiment.
- Let B be the event that the first number thrown is no larger than 3, and let C be the event that the sum of the two numbers thrown equals 6. Find the probabilities of B and C , and the conditional probabilities of C given B , and of B given C .

a. $S = \{1, 2, 3, 4, 5, 6\}$

b. $B = \{1, 2, 3\}$ for first number

$C = \{\text{sum} : \text{sum} = 6\}$

$$\therefore P(B) = \frac{3}{6} = \frac{1}{2}$$

$$P(C) = \frac{5}{36} = \frac{1}{7.2}$$

$$\therefore P(B \cap C) = P(C \cap B) = \frac{2}{36} = \frac{1}{18}$$

$$\therefore P(C|B) = \frac{P(C \cap B)}{P(B)} = \frac{1}{6}$$

$$P(B|C) = \frac{P(B \cap C)}{P(C)} = \frac{1}{2} \cdot \frac{3}{5}$$

3. Cables a and b can support a load W . Normally a supports the load, but if a develops a fault then b comes into operation. Let A denote the event that a fails and B denotes the event that b fails. Write down the event that the load remains supported.

A : a fails

B : b fails

$\therefore$ load remains supported is

$$\bar{A} \cup \bar{B}$$

4. Events A , B and C have probabilities 0.7, 0.8 and 0.9 respectively. In addition $P(A \cap B) = 0.6$, $P(B \cap C) = 0.7$, $P(A \cap C) = 0.6$, $P(A \cap B \cap C) = 0.5$. Calculate:

- $P(A \cap \bar{B})$
- $P(A \cup B)$
- $P(A \cap B \cap \bar{C})$
- $P(A \cup B \cup C)$

$$a. P(A \cap \bar{B}) = P(A) - P(A \cap B) = 0.1$$

$$b. P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.9$$

$$c. P(A \cap B \cap \bar{C}) = P((A \cap B) \cap \bar{C})$$

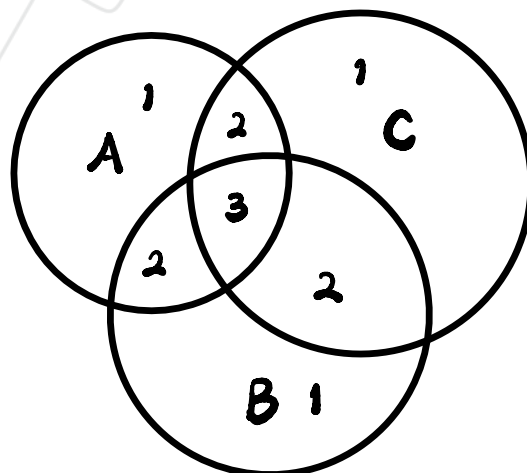
$$= P(A \cap B) - P(A \cap B \cap C) = 0.1$$

$$d. P(A \cup B \cup C)$$

$$= P(A) + P(B) + P(C)$$

$$- P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

$$= 1$$



5. Prove statements 2 - 4 of Propositions 1.1 and 1.2.

1.1 1. $A \cup \phi = A$

$$x \in A \cup \phi \quad \therefore x \in A \text{ or } x \in \phi \quad \therefore x \in A$$

2. $A \cup A = A$

$$x \in A \cup A \quad \therefore x \in A \text{ or } x \in A \quad \therefore x \in A$$

3. $A \cup S = S$

$$x \in A \cup S \quad \therefore x \in A \text{ or } x \in S \quad \therefore x \in S$$

4. $A \cup B = B \cup A$

$$x \in A \cup B \quad \therefore x \in A \text{ or } x \in B \quad \therefore x \in B \cup A$$

1.2 1. $(\bar{A}) = A$

$$x \in S \text{ and } x \notin \bar{A} \quad \therefore x \in S \text{ and } x \in A \quad \therefore x \in A$$

2. $\bar{\phi} = S$

$$x \in S \text{ and } x \notin \phi \quad \therefore x \in S$$

3. $A \cup \bar{A} = S$

$$x \in A \cup \bar{A} \quad \therefore x \in A \text{ or } x \in \bar{A} \quad \therefore x \in S$$

4. $A \cap \bar{A} = \phi$

$$x \in A \cap \bar{A} \quad \therefore x \in A \text{ and } x \in \bar{A} \quad \therefore x \in \phi$$

6. Prove statements (ii) and (iv) of Proposition 2.1.

2.1 1. $P(A|B) \geq 0$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\because P(A \cap B) \geq 0 \text{ and } P(B) \geq 0 \quad \therefore P(A|B) \geq 0$$

2. $P(S|B) = 1$

$$P(S|B) = \frac{P(S \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

3. If A and B are disjoint, $P(A \cup B|C) = P(A|C) + P(B|C)$

$$\begin{aligned} P(A \cup B|C) &= \frac{P((A \cup B) \cap C)}{P(C)} = \frac{P(A \cap C) + P(B \cap C)}{P(C)} \\ &= P(A|C) + P(B|C) \end{aligned}$$

4. $P(\bar{A}|B) = 1 - P(A|B)$

$$\begin{aligned} P(\bar{A}|B) &= \frac{P(\bar{A} \cap B)}{P(B)} = \frac{P(B) - P(A \cap B)}{P(B)} \\ &= 1 - P(A|B) \end{aligned}$$

7. An electrical device, D , consists of two independent components, which are connected in parallel. Each component functions with probability 0.7. What is the probability that a current can flow through the device when it is connected to a battery? A new device, D' , is obtained by connecting two independent devices of the same type as D above, in series. What is the probability that a current can flow through this new device when it is connected to a battery?

Probability of current can flow through the component :

$$P = 1 - P(\bar{D} | \bar{D}) P(\bar{D}) = 1 - P(\bar{D}) P(\bar{D}) = 0.91$$

Probability of current can flow through new device :

$$P' = P \cap P = P \cdot P = 0.8281$$

8. In question 3, the events A and B are independent and each has probability 0.01. Find the probability that W remains supported.

$$\begin{aligned}P(\bar{A} \cup \bar{B}) &= 1 - P(A \cap B) \\&= 1 - [P(A) + P(B) - P(A \cap B)] \\&= 1 - [P(A) + P(B) - P(A)P(B)] \\&= 1 - (0.01 + 0.01 - 0.01 \cdot 0.01) \\&= 0.9801\end{aligned}$$

$$\begin{aligned}&= P(\overline{A \cap B}) \\&= 1 - P(A \cap B) \\&= 1 - P(A)P(B) \\&= 1 - (0.01)^2 \\&= 0.9999\end{aligned}$$

9. An aircraft is built in 2 main stages. The probability that stage 1 is completed on time is 0.8. If stage 1 is completed on time the probability that the materials required for stage 2 will be available is 0.9. If stage 1 is not completed on time the probability that the materials required for stage 2 will be available is 0.3.
- Find the probability that the materials for stage 2 will be available.
 - If the materials for stage 2 are available, what is the probability that stage 1 was completed on time?

$$a. P(1) = 0.8$$

$$P(2|1) = 0.9$$

$$P(2|\bar{1}) = 0.3$$

$$\therefore P(2) = P(2|1)P(1) + P(2|\bar{1})P(\bar{1})$$

$$= 0.9 \cdot 0.8 + 0.3 \cdot 0.2$$

$$= 0.78$$

$$b. P(1|2) = \frac{P(1 \cap 2)}{P(2)}$$

$$= \frac{P(2|1)P(1)}{P(2|1)P(1) + P(2|\bar{1})P(\bar{1})}$$

$$= \frac{0.9 \cdot 0.8}{0.78}$$

$$= 0.92307$$

10. A navigational device, D , on an aircraft functions (event F) with probability 0.95. It is subjected to 2 independent diagnostic tests, A and B , to check for faults. If either (or both) test gives a positive result, then D is declared unusable (event U). However, A and B are not foolproof. If D is in fact functional, A and B will give positive results with probabilities 0.15 and 0.20 respectively. If D is in fact non-functional, A and B will give positive results with probabilities 0.95 and 0.80 respectively.

Let $A +$ and $B +$ denote the events that A and B give positive results, respectively.

- Write down the event U in terms of $A +$ and $B +$.
- Calculate $P(U|F)$ and $P(U|\bar{F})$.
- What is the probability that D is declared unusable?
- Suppose now that D is tested and as a result declared unusable. What is the probability that in fact it does not function?

$$P(F) = 0.95 \quad \therefore P(\bar{F}) = 0.05$$

$$P(A+|F) = 0.15 \quad \therefore P(A-|F) = 0.85$$

$$P(B+|F) = 0.20 \quad \therefore P(B-|F) = 0.8$$

$$P(A+|\bar{F}) = 0.95 \quad \therefore P(A-|\bar{F}) = 0.05$$

$$P(B+|\bar{F}) = 0.80 \quad \therefore P(B-|\bar{F}) = 0.2$$

$$a. U = (A+ \overset{U}{\cap} B+) \quad \cancel{U = (A- \cap B-)}$$

$$b. P(U|F) = P(\overset{A+ \cup B+}{(A+ \cap B+) \cup (A- \cap B-)} | F)$$

$$= \cancel{P((A+ \cap B+) | F) + P((A- \cap B-) | F)}$$

$$= \cancel{P(A+|F)P(B+|F) + P(A-|F)P(B-|F)}$$

$$= \cancel{0.71} \quad 0.32$$

$$P(U|\bar{F}) = \frac{P(A+1|\bar{F}) + P(B+1|\bar{F}) - P(A+1|\bar{F})P(B+1|\bar{F})}{P(A+1|\bar{F}) + P(B+1|\bar{F}) - P(A+1|\bar{F})P(B+1|\bar{F})}$$

$$= \frac{0.77 + 0.99 - 0.77 \cdot 0.99}{0.77 + 0.99 - 0.77 \cdot 0.99}$$

$$c. P(U) = P(U|F)P(F) + P(U|\bar{F})P(\bar{F})$$

$$= 0.713 + 0.3535$$

$$d. P(\bar{F}|U) = \frac{P(U|\bar{F})P(\bar{F})}{P(U|\bar{F})P(\bar{F}) + P(U|F)P(F)}$$

$$= \frac{0.053497}{0.053497 + 0.1490}$$